

CSE 144 Final Project

Aashna Shimkhada

Network Backbone

- EfficientNet-B3 pretrained on ImageNet (~12M params)
- Why B3? Compound scaling — strong accuracy-to-compute ratio
- Custom 2-layer classifier head: Dropout(0.4) → Linear(1536→512) → ReLU → Dropout(0.3) → Linear(512→100)
- Gradual unfreezing: top 4 backbone blocks in Phase 2, then 6 blocks at epoch 10

Training Pipeline

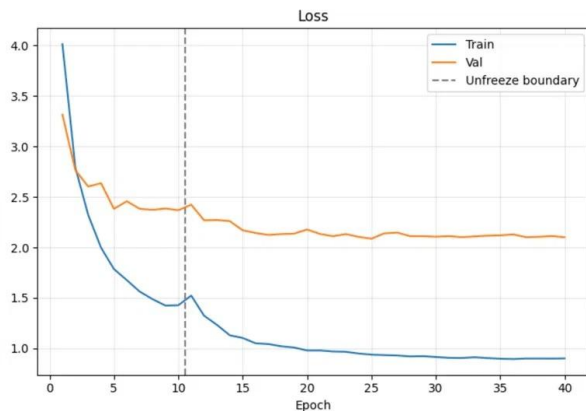
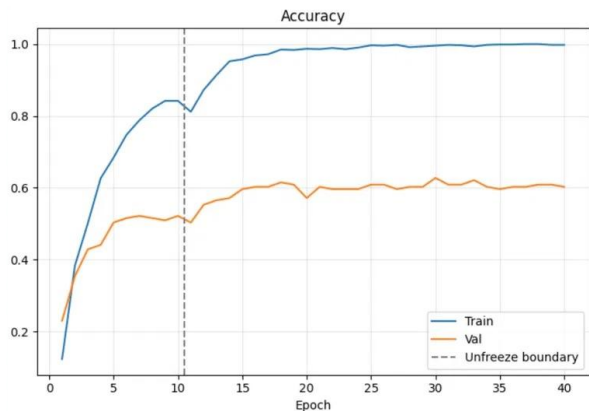
- Phase 1 (10 epochs): Head-only, backbone fully frozen, $LR=3e-3$
- Phase 2 (up to 30 epochs): Gradual unfreeze — differential LRs
 - Backbone: $LR=1e-4$ | Head: $LR=1e-3$ ($10\times$ ratio)
 - Epoch 10: unfreeze 2 more blocks at $0.5\times$ backbone LR
- Early stopping (patience=12) saves best checkpoint automatically
- Scheduler: CosineAnnealingWarmRestarts ($T_0=10$) — helps escape local minima

Hyperparameters

- Augmentation: random crop, horizontal flip, rotation $\pm 15^\circ$, color jitter, random erasing
- All decisions made on 15% held-out val split

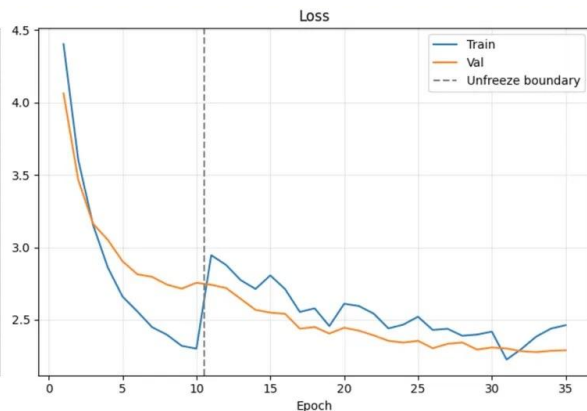
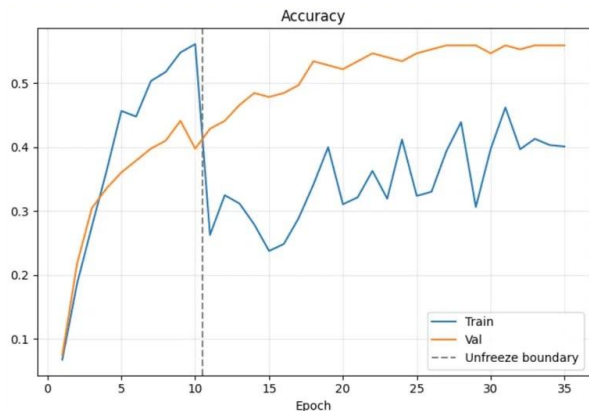
Parameter	Value
Image Size	224 x 224
Batch Size	32
Phase 1 LR	3e-3
Backbone LR (Phase 2)	1e-4
Head LR (Phase 2)	1e-3
Weight Decay	1e-4
Mixup α	0.4
Label Smoothing	0.1
Early Stopping Patience	12

Current Results & Training Curves



Plot 1 (v1) — Severe overfitting:

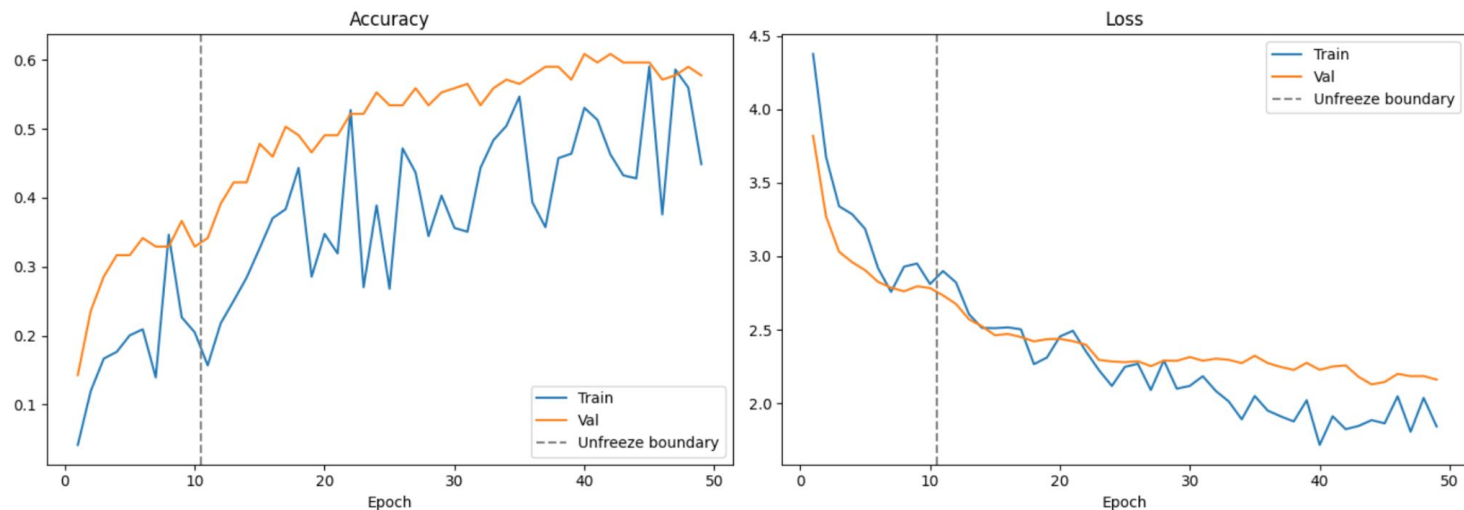
- Train acc $\rightarrow$ 100%, Val acc $\rightarrow$ ~60% | 40-epoch gap never closes
- Val loss flat while train loss falls $\rightarrow$ model memorizing training set



Plot 2 (v2 — my overcorrection):

- Added Mixup $\alpha=0.4$, stronger dropout, LR=5e-5
- Train acc (~40%) < Val acc (~57%) — Mixup artifact: train measured on blended images
- Val loss still slowly declining $\rightarrow$ more epochs likely to help

Current Results & Training Curves (cont.)



Plot 3 (v3 — current run, 50 epochs):

- Val acc peaks at ~61% and holds — model has converged
- Train acc noisy from Mixup but trending up, gap with val small and closing
- Val loss still slowly declining — a few more epochs may squeeze out 1–2% more

Next Steps

- A few more epochs — val loss still slowly declining; ~55 epochs likely squeezes 1–2% more
- Lower Mixup α ($0.4 \rightarrow 0.2$) — would make the training curve less noisy and easier to diagnose
- CutMix — blends image patches instead of pixels; generally stronger than Mixup alone
- Unfreeze full backbone — currently only top 6 of 9 blocks; unfreezing all at lower LR may help
- Ensemble — 3 runs $\times$ different seeds typically adds 2–3% at no architecture cost